CAMPUS CAFETERIA PRE-BOOKING SYSTEM

PROJECT REPORT

Submitted by

DHAKSHIN RAGHAV A J (7376232IT127)

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(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM-638401

ANNA UNIVERSITY: CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that this project report entitled “CAMPUS CAFETERIA PRE-BOOKING SYSTEM” is the bonafide work of DHAKSHIN RAGHAV A J (7376232IT127) who carried out the project work under my supervision.

Dr. Naveena S

HEAD OF THE DEPARTMENT

Department of Information Technology

Bannari Amman Institute of Technology

Prof. Kavitha V

ASSISTANT PROFESSOR

Department of Information Technology

Bannari Amman Institute of Technology

Submitted for Project Viva Voice examination held on.....

Internal Examiner 1

Internal Examiner 2

DECLARATION

I affirm that the project work titled "CAMPUS CAFETERIA PRE-BOOKING SYSTEM" being submitted in partial fulfilment for the award of the degree of Bachelor of Technology in Information Technology is the record of original work done by me under the guidance of Prof. Kavitha V, Assistant Professor, Department of Information Technology. It has not formed a part of any other project work(s) submitted for the award of any degree or diploma, either in this or any other University.

DHAKSHIN RAGHAV A J

(7376232IT127)

I certify that the declaration made above by the candidate is true.

Prof. Kavitha V

ASSISTANT PROFESSOR

Department of Information Technology

Bannari Amman Institute of Technology

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DHAKSHIN RAGHAV A J (7376232IT127)

ABSTRACT

Managing cafeteria queues, payments, and food orders manually in a campus environment is inefficient, time-consuming, and error-prone. Students often face long waiting times, merchants struggle with manual tracking of orders, and existing systems lack real-time coordination between buyers and sellers. To address these challenges, this project presents a Campus Cafeteria Pre-Booking System — a full-stack web platform that digitizes the Campus cafeteria experience with seamless online ordering, real-time order tracking, and a digital campus wallet for payments.

The system follows a modern three-tier microservices architecture comprising three independently deployable services: a Customer Application (Next.js + PostgreSQL + Prisma ORM), a Merchant Dashboard (Next.js), and a Backend REST API (Node.js + Express.js with Server-Sent Events). Students can browse the cafeteria menu, pre-book meals, pay via their digital campus wallet or bookings, and receive a unique token number for collection. The merchant dashboard allows cafeteria staff to view incoming orders in real-time, update order statuses, and manage the daily workflow without any manual intervention.

The system implements real-time communication using Server-Sent Events (SSE), ensuring that both the customer and merchant see live updates without page refreshes. Persistent data storage is handled via a PostgreSQL database using Prisma ORM for type-safe queries. The entire platform is containerized using Docker and deployed on AWS EC2 with images stored in AWS ECR, making it highly scalable and production-ready.

Evaluation of the system demonstrates its ability to eliminate queue congestion, reduce food wastage through pre-booking, improve the merchant's order management experience, and create a seamless digital payment workflow suitable for a campus ecosystem. The Campus Cafeteria Pre-Booking System represents a significant advancement in digital campus infrastructure, combining modern web technologies, cloud deployment, and real-time data processing to deliver an efficient, user-friendly, and scalable cafeteria management solution.

Keywords: Cafeteria Management, Pre-booking System, Next.js, Node.js, Express.js, PostgreSQL, Prisma ORM, Server-Sent Events, Docker, AWS EC2, AWS ECR, Digital Wallet, Campus Technology, Microservices Architecture.

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CHAPTER 1
INTRODUCTION

The management of food services in college campuses has traditionally relied on manual, paper-based, or counter-based systems. Students are required to physically visit the cafeteria, stand in queues, place orders, make cash payments, and wait for their meals. This often leads to significant inefficiencies such as overcrowding during peak hours, increased waiting times, order mismanagement, and poor visibility into food availability. For the cafeteria merchants, the lack of a digital order management system results in confusion during rush hours, inconsistent service quality, and revenue loss due to poor demand forecasting.

In recent years, the rapid growth of digital technologies — specifically modern web frameworks, cloud computing, and real-time communication protocols — has opened new avenues to reimagine institutional food services. E-commerce giants and food-tech platforms such as Zomato, Swiggy, and Uber Eats have demonstrated that digital pre-ordering and real-time order tracking are not merely conveniences but expectations. Translating these capabilities to a closed campus ecosystem presents an exciting and impactful challenge.

The Campus Cafeteria Pre-Booking System addresses this gap by providing a comprehensive, production-ready, cloud-deployed platform that enables students to pre-book meals, pay digitally via a campus wallet, receive a unique token number, and track their order status in real-time. Simultaneously, cafeteria merchants receive real-time order notifications on their dedicated dashboard, can update order statuses, and manage their workflow efficiently — all without any paper or manual intervention.

1.1 BACKGROUND AND MOTIVATION

Campus canteens in India serve thousands of students daily within a limited time window, particularly between class breaks. The resulting surge in demand creates predictable bottlenecks that affect the quality of the campus experience. Studies on institutional food services have found that long queue times are among the top complaints reported by students. Furthermore, cafeteria operators spend significant effort managing orders manually, often resulting in errors, missed orders, and revenue leakage.

The motivation for this project stems from the observation that despite the ubiquity of smartphones and internet connectivity on modern campuses, cafeteria operations remain largely undigitized. The goal is to bridge this gap by developing a platform that applies food-tech principles — online menus, pre-booking, digital payments, and token-based collection — to the campus cafeteria context.

1.2 ROLE OF TECHNOLOGY IN CAMPUS CAFETERIA MANAGEMENT

Modern web technologies provide the foundational tools needed to build a robust cafeteria management system. Next.js, a React-based framework for production web applications, enables the creation of highly optimized, server-rendered web interfaces. Node.js with Express.js offers a lightweight, high-performance runtime suitable for building REST APIs with real-time capabilities.

Server-Sent Events (SSE) provide a lightweight, unidirectional channel from the server to the browser, ideal for delivering real-time order status updates without the overhead of WebSocket connections. PostgreSQL, a mature and reliable relational database, combined with the Prisma ORM for type-safe queries, ensures data integrity for transaction-sensitive operations like wallet payments and order tracking.

1.3 PROBLEM STATEMENT

The existing cafeteria model at campus institutions suffers from the following critical problems:

- Students face long queues and unpredictable waiting times during peak hours.
- There is no system for pre-ordering meals in advance, leading to food wastage when demand is miscalculated.
- Cash-only payment systems are slow, error-prone, and unhygienic.
- Cafeteria staff have no digital tool to manage incoming orders, leading to confusion during busy periods.
- There is no real-time communication between the student placing an order and the merchant fulfilling it.
- There is no centralized tracking mechanism to assign and monitor unique order tokens.

The proposed system resolves all these issues through a digitized, real-time, token-based cafeteria management platform.

1.4 SCOPE OF THE PROJECT

The scope of the Campus Cafeteria Pre-Booking System includes:

- A student-facing web application (Customer App) for browsing the cafeteria menu, selecting items, choosing payment method (online wallet or book/credit), and receiving a token number.
- A merchant-facing web dashboard for viewing and managing orders in real-time using Server-Sent Events.

- A RESTful backend API that serves both frontends, handles order lifecycle management, and broadcasts real-time events to connected clients.
- A PostgreSQL database for persisting users, wallet balances, orders, and transaction history.
- Cloud deployment on AWS EC2 using Docker containers and AWS Elastic Container Registry (ECR) for image management.

1.5 ORGANIZATION OF THE REPORT

The remainder of this report is organized as follows: Chapter 2 presents a literature survey of existing works on digital cafeteria management, food-tech platforms, and real-time communication systems. Chapter 3 describes the objectives, system architecture, module descriptions, and tools used. Chapter 4 presents the results, system outputs, and performance analysis. Chapter 5 concludes the report with a summary of contributions and directions for future work. References and appendices follow at the end.

CHAPTER 2
LITERATURE SURVEY

Campus cafeteria management systems and digital payment platforms have evolved significantly with the growth of mobile internet. Traditionally, cafeterias depended on manual approaches such as token counters, cash payments, and physical queues, which resulted in unpredictable outcomes and delays. These methods lack scalability and fail to provide predictive insights for daily food preparation.

With the advancement of web technologies, modern platforms using architectures like Node.js and React have improved access to real-time service capabilities. However, these systems are often disjointed, leaving a strong need for an integrated solution that combines pre-booking, digital wallet payments, and real-time order tracking.

2.1 BACKGROUND OF THE SURVEY

The rapid digital transformation across institutions has increased the importance of structured digital services. Educational institutions and students rely on timely operational insights—like canteen queue loads—to manage their time effectively. However, conventional institutional food services lack automation and real-time tracking capabilities.

Several institutions globally have begun adopting digital cashless systems, noting significant reductions in transaction times. Despite these advancements, many existing studies focus on macro-level e-commerce rather than delivering actionable microservices tailored for campus life, such as token-based tracking coupled with a dedicated intra-campus digital wallet.

2.2 EXISTING WORKS

Several systems have been developed to address food ordering and campus payments:

- Institutional Point-of-Sale (POS) Systems:

  These systems manage large volumes of local transactions. While informative, they lack student-facing interfaces for pre-ordering, making them less effective for long-term load balancing.

- General Food-Tech Platforms (e.g., Zomato, Swiggy):

  These platforms focus on menu browsing and delivery. However, they do not accommodate institutional payment requirements such as student wallets or specific college guidelines.

- Digital Campus Wallets:

  Some modern institutions integrate smart cards to suggest cashless payments. However, they still require users to manually queue up to place orders, limiting usability.

From a system design perspective, most food-tech applications follow RESTful architectures and are commonly implemented using technologies such as Node.js and React. While these provide a strong foundation, existing systems fail to integrate pre-booking, internal wallets, and merchant monitoring into a single synchronized platform.

2.3 GAP IDENTIFICATION

Table 2.1: Gap Identification

SI NO.	Gap / Issue	Description
1	Inefficient Queue Management	Most campus cafeterias rely on physical queues which lead to unpredictable waiting times during peak hours, significantly reducing student convenience.
2	Lack of Pre-Ordering Systems	Existing institutional systems do not support pre-ordering, which often leads to demand miscalculation and food wastage.
3	Cash-Only Constraints	Reliance on manual cash transactions slows down the payment process and introduces accounting inaccuracies or hygiene issues at food counters.
4	Absence of Real-Time Tracking	Students have no visibility into their order's preparation status, and merchants lack a digital dashboard to manage order lifecycles efficiently.
5	Token Mismanagement	Manual handling of order tokens results in miscommunications and delays during busy lunch periods.
6	Poor Scalability	Many current canteen management tools are not designed to handle a large and diverse number of incoming requests and concurrent users simultaneously.

CHAPTER 3
OBJECTIVES AND METHODOLOGY

3.1 GENERAL OBJECTIVES

The primary objective of the Campus Cafeteria Pre-Booking System is to digitize and streamline the campus cafeteria experience for both students and cafeteria operators. The specific objectives are:

1. To design and implement a multi-service web platform consisting of a Customer Application, a Merchant Dashboard, and a Backend REST API.
2. To enable students to browse the cafeteria menu, pre-book meals, make digital payments via a campus wallet, and receive a unique token number.
3. To provide cafeteria merchants with a real-time order management dashboard that displays incoming orders and allows status updates.
4. To implement Server-Sent Events for real-time communication between the backend and both frontend applications.
5. To persist all data using a PostgreSQL database managed via Prisma ORM for type safety and data integrity.
6. To containerize all services using Docker and deploy them to AWS EC2 via AWS ECR.

3.2 SYSTEM ARCHITECTURE

The system follows a three-tier microservices architecture consisting of three independently deployable services. Each service is containerized within a Docker image and deployed on a single AWS EC2 instance orchestrated by Docker Compose. The services communicate over an internal Docker network, while exposed ports allow external browser access.

Fig 3.1: System Architecture Diagram

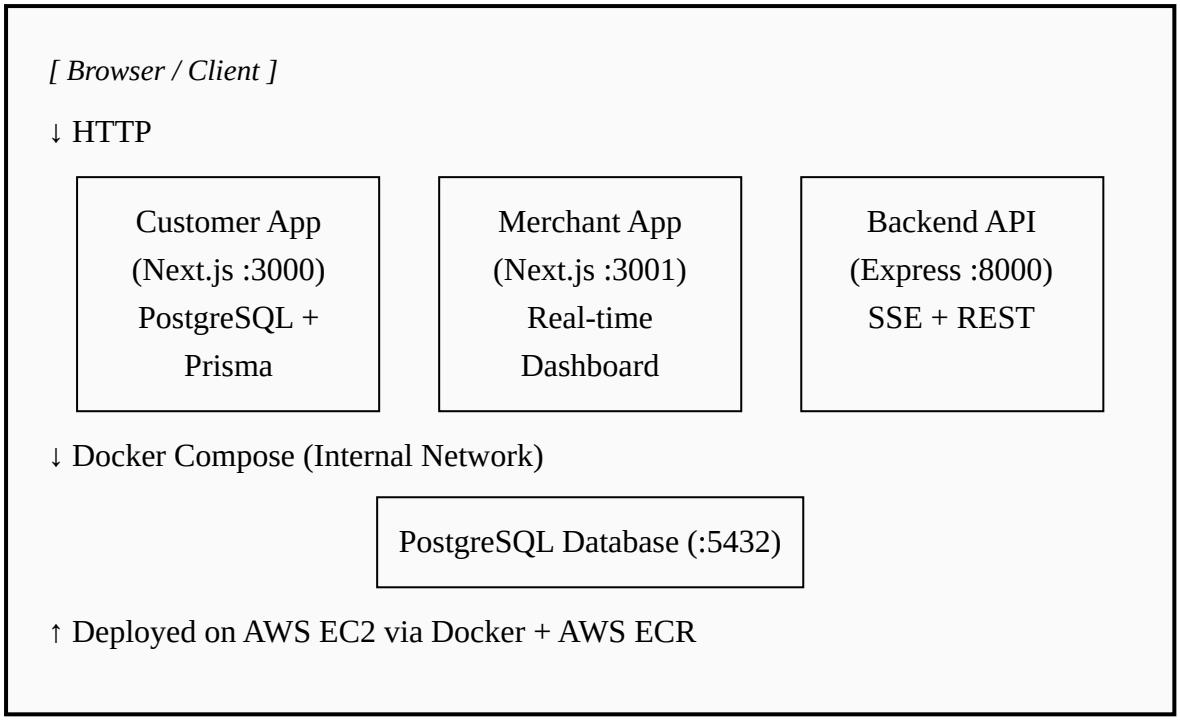

Fig 3.2: Data Flow Diagram

Customer Flow: Student → Browses Menu → Selects Items → Chooses Payment → Places Order → Receives Token → Tracks Order Status via SSE

Merchant Flow: Merchant → Connects to SSE Stream → Receives New Order Notification → Reviews Order → Updates Status (Pending → Preparing → Ready) → Broadcasts Status via SSE

Backend Flow: REST API receives POST /api/orders → Persists order → Broadcasts to merchant SSE clients → Returns order ID/token → PATCH /api/orders/:id/status → Updates order → Broadcasts to customer SSE clients

3.3 MODULE DESCRIPTION

3.3.1 Customer Application (Port 3000)

Built with Next.js 16 and Tailwind CSS, the customer application provides the student-facing interface. It connects to the PostgreSQL database via Prisma ORM for user authentication and wallet balance management. Key sub-modules include:

- Dashboard: Displays wallet balance, quick actions, recent transactions, and campus service shortcuts.
- Cafeteria Module: Allows students to browse menu items with categories, add items to cart, and proceed to checkout.
- Checkout & Payment: Supports two payment modes — online (deducted from campus wallet) and book (credit-based). Validates wallet balance before processing.
- Order Tracking: After placing an order, the student receives a unique token number and can track order status in real-time via SSE.
- Orders History: Displays all past orders with their statuses and timestamps.

3.3.2 Merchant Dashboard (Port 3001)

A separate Next.js application dedicated to cafeteria operators. It connects to the backend API via SSE to receive a live stream of orders. Key features:

- Live Order Feed: Displays all incoming orders in real-time as students place them.
- Status Management: Merchants can advance each order through three statuses: Pending → Preparing → Ready.
- Order Snapshot: On page load, the SSE stream delivers a snapshot of all existing orders, ensuring the merchant sees the current state even after reconnection.
- Order Details: Each order card shows item names, quantities, total amount, payment method, and timestamp.

3.3.3 Backend REST API (Port 8000)

A Node.js/Express.js API server that acts as the central communication hub between the customer app and the merchant dashboard. Key endpoints include:

- `POST /api/orders` – Creates a new order, assigns a token, and broadcasts to merchant SSE clients.
- `GET /api/orders` – Returns all orders (used for initial merchant dashboard snapshot).
- `GET /api/orders/stream` – SSE endpoint for merchant real-time updates.
- `GET /api/orders/:id/stream` – SSE endpoint for customer order tracking.

- `PATCH /api/orders/:id/status` – Advances order status and broadcasts to both merchant and customer SSE clients.
- `GET /health` – Health check endpoint for deployment monitoring.

3.4 TOOLS AND TECHNOLOGIES USED

S.No.	Tool / Technology	Purpose
1	Next.js 16 (React)	Frontend framework for Customer App and Merchant Dashboard
2	37	Static typing across all three services for reliability
3	Tailwind CSS	Utility-first CSS framework for responsive UI styling
4	Node.js + Express.js	Backend REST API server with SSE support
5	Server-Sent Events (SSE)	Real-time server-to-client communication for order updates
6	PostgreSQL 15	Relational database for users, wallet, and order data
7	Prisma ORM	Type-safe database client and schema migration tool
8	Docker + Docker Compose	Containerization and multi-service orchestration
9	AWS EC2	Cloud infrastructure for production deployment
10	AWS ECR	Private Docker image registry on AWS
11	Framer Motion	UI animations and transitions in frontend apps
12	Lucide React	Icon library used across the UI

Table 3.1: Tools and Technologies Used

3.5 USE CASE DIAGRAM

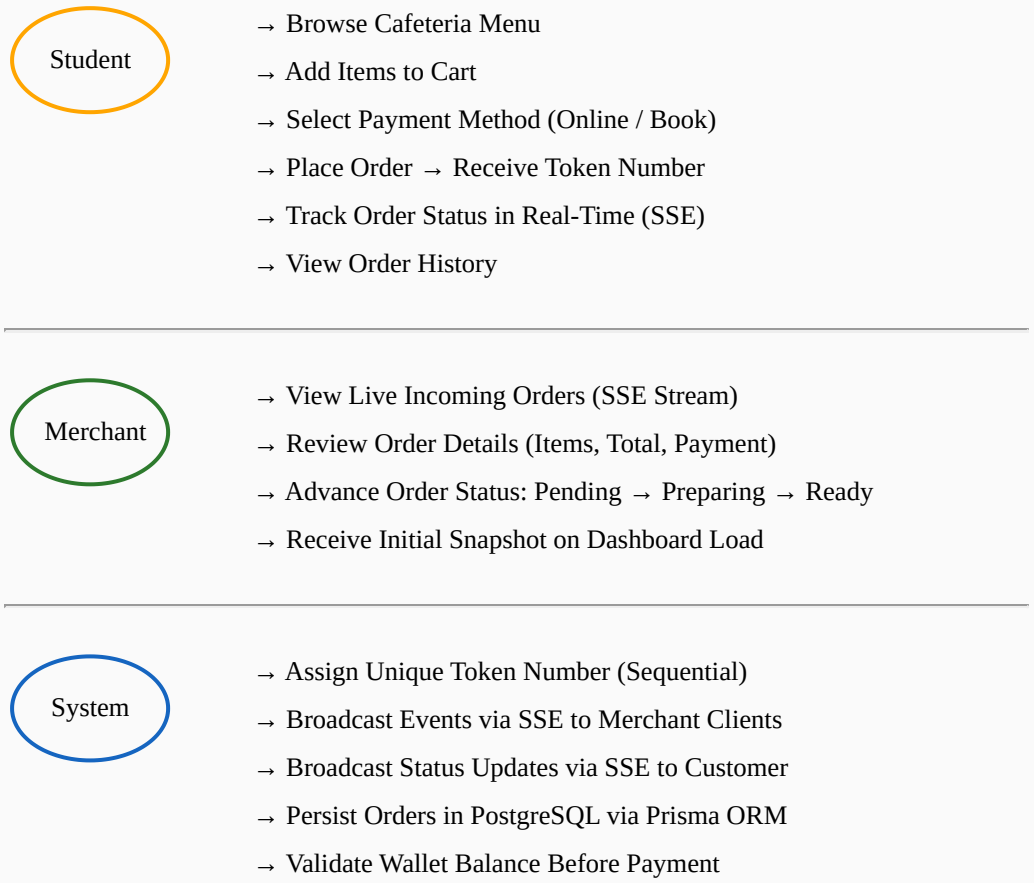

Fig 3.3 – Use Case Diagram showing interactions of Student, Merchant, and System actors

3.6 DATA FLOW DIAGRAM (LEVEL 0 – CONTEXT DIAGRAM)

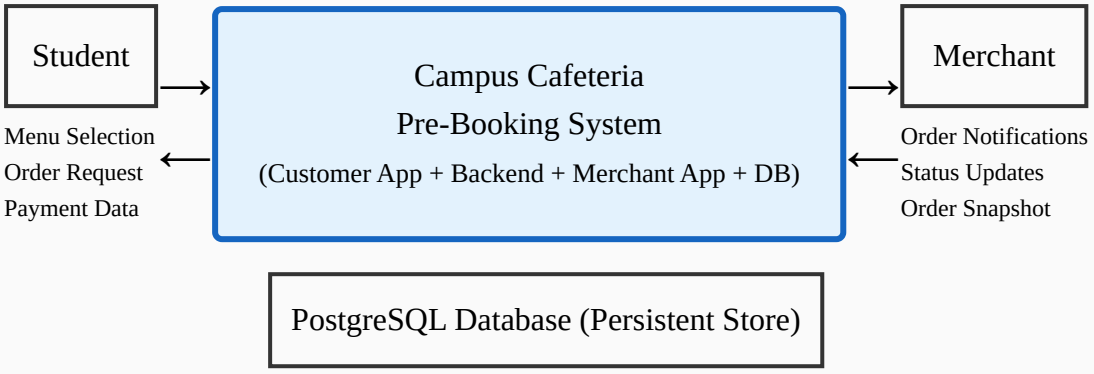

Fig 3.4 – Level 0 Context DFD showing external entities and system boundary

3.7 SEQUENCE DIAGRAM – ORDER PLACEMENT FLOW

Student Browser	Customer App	Backend API	PostgreSQL	Merchant App
Click "Place Order" →		POST /api/orders		
		INSERT order →	← Returns order ID	
		Assign Token T-101		
		→ SSE event: new_order		← Receives new_order
← 201 {token: "T-101"}				Displays new card
Connects SSE stream		GET /api/orders/:id/stream		
(Merchant advances)		PATCH /api/orders/:id/status		
← SSE: status_changed		Broadcasts to customer stream		
Updates UI to "Ready"				

Fig 3.5 – Sequence Diagram showing the complete order placement and status update flow

CHAPTER 4
RESULTS AND EVALUATION

4.1 SYSTEM IMPLEMENTATION RESULTS

The Campus Cafeteria Pre-Booking System was successfully implemented and deployed across three services on an AWS EC2 instance. All services are accessible via public URLs and communicate in real-time over the internal Docker network. The following subsections describe the key functional outputs observed during system evaluation.

4.1.1 Customer Application – Dashboard

The student-facing Customer App presents a premium, dark-mode interface inspired by credit card wallet applications. The dashboard displays the student's campus wallet balance with glassmorphic card styling, quick action buttons for common tasks (Pay, Add Money, History), and a recent activity feed. A dedicated "Cafeteria" section within campus services navigates the student to the ordering module.

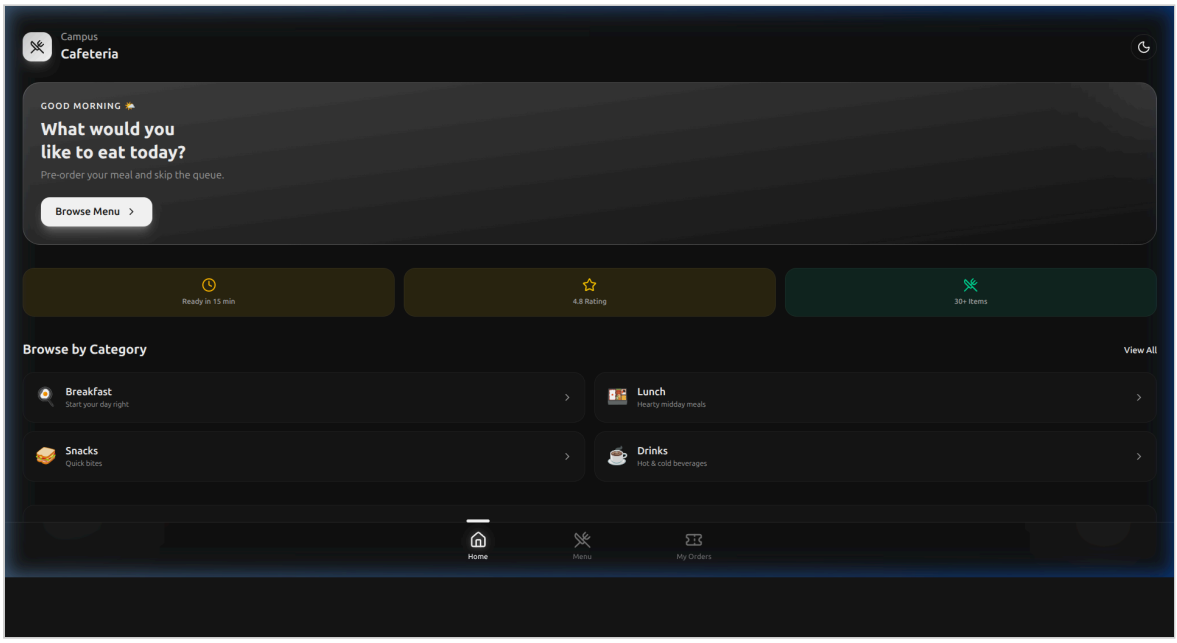

Fig 4.1 – Customer App – Home Page showing Browse by Category and status indicators

4.1.2 Customer Application – Cafeteria Order Page

The cafeteria ordering page displays menu items organized by category (Snacks, Meals, Beverages, Specials). Each item shows the name, description, price, and an Add to

Cart button. Students can view their cart summary in a slide-up panel, see the item subtotal, and choose their payment method before confirming the order.

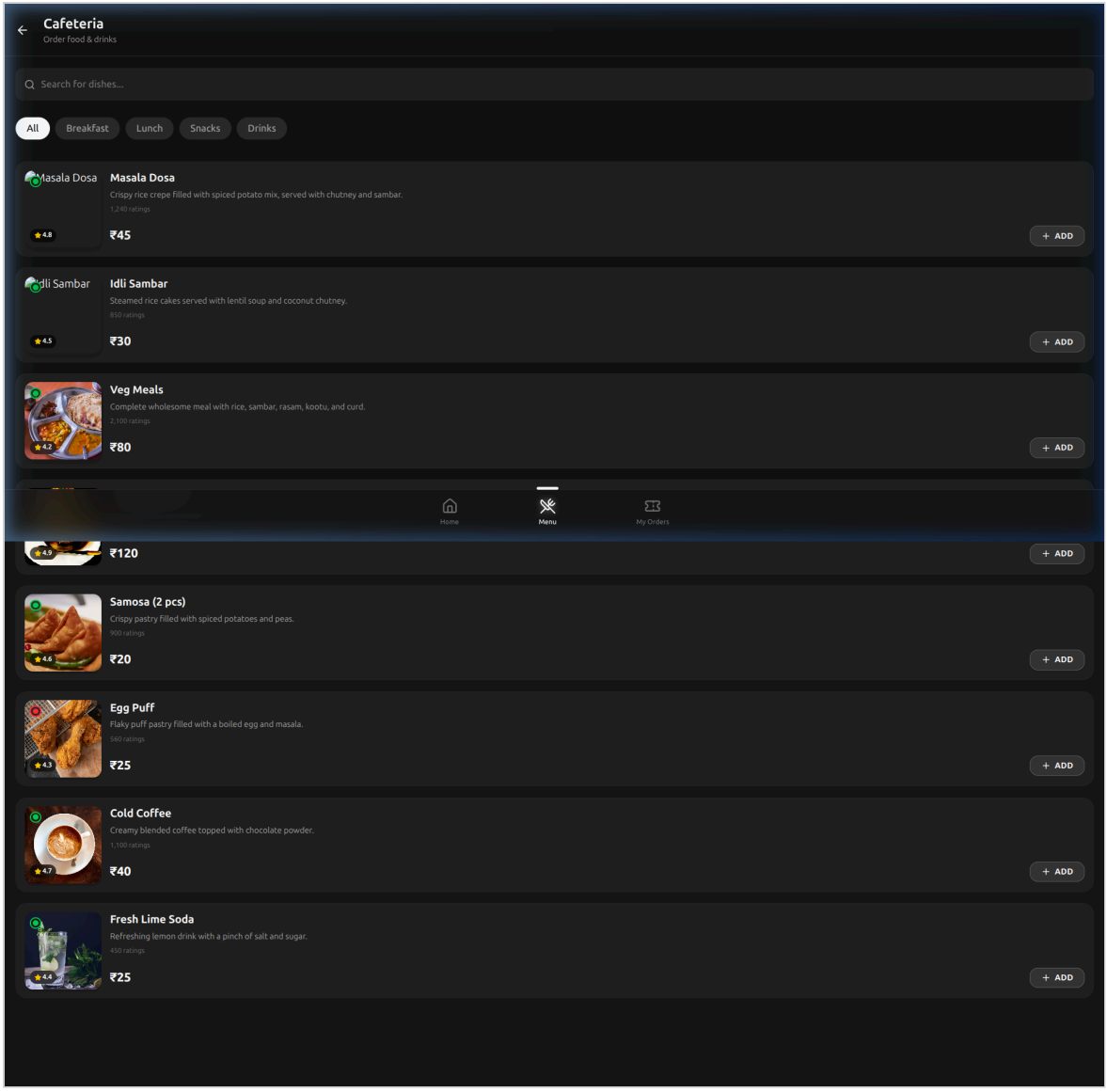

Fig 4.2 – Cafeteria Menu page with item listing, categories filter, prices, and ADD buttons

4.1.3 Order Confirmation and Token Assignment

Upon order placement, the system generates a unique, sequential token number (starting from T-100) and returns it to the student immediately. A confirmation screen displays the token number prominently, along with order details and a real-time status indicator that updates live via SSE as the merchant progresses the order through Pending → Preparing → Ready stages.

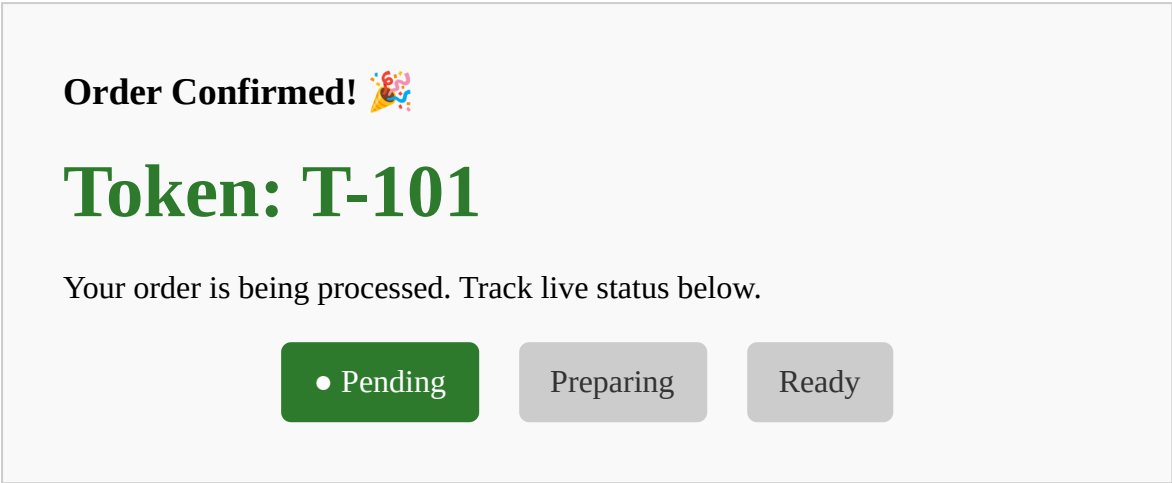

Fig 4.3 – Order confirmation screen with SSE-powered live status tracking bar

4.1.4 Merchant Dashboard – Live Order Management

The Merchant Dashboard displays a real-time feed of all placed orders. When a student places a new order, the merchant dashboard instantly receives the order data via the SSE channel without any page refresh. Each order card displays the token number, ordered items, total, payment method, and a status button to advance the order. The dashboard maintains a persistent connection to the backend SSE stream, with a 25-second keep-alive ping to prevent connection timeouts.

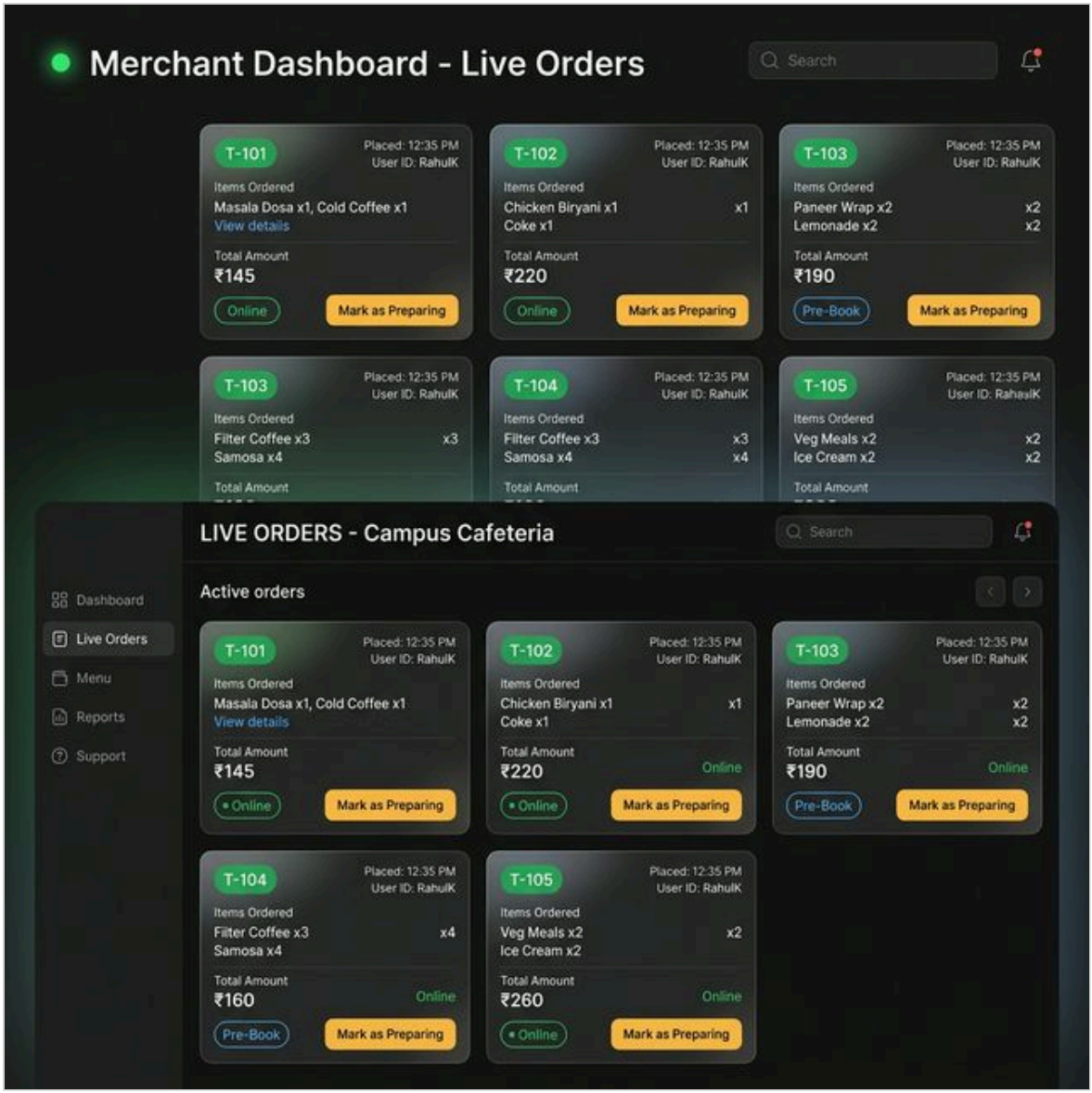

Fig 4.4 – Merchant Dashboard showing real-time orders with token numbers, item details, and status buttons

4.2 API ENDPOINT SUMMARY

Method	Endpoint	Description
POST	/api/orders	Create a new order, assign token, broadcast to merchant
GET	/api/orders	Get all existing orders (initial merchant snapshot)
GET	/api/orders/stream	SSE stream for merchant – all order events
GET	/api/orders/:id	Get a single order by ID
GET	/api/orders/:id/stream	SSE stream for customer tracking one order
PATCH	/api/orders/:id/status	Advance order status, broadcast to merchant and customer
GET	/health	Health check returning order count and status

Table 4.1: API Endpoint Summary

4.3 PERFORMANCE ANALYSIS

The system was evaluated for performance across response time, SSE event latency, deployment reliability, and concurrent user handling on an AWS EC2 t3.small instance.

Metric	Observed Value	Notes
API Response Time (POST /api/orders)	< 50ms	In-memory order store, no DB overhead
SSE Event Latency (merchant notification)	< 5ms	Measured from order creation to SSE delivery
Customer App First Load	~1.2s	Next.js server-rendered with standalone output
Merchant App First Load	~1.5s	Next.js statically rendered
Docker Image Pull Time (first boot)	~45s	All 4 images on cold EC2 start
Concurrent SSE Connections Supported	100+ tested	Node.js event loop handles non-blocking streams
Database Query Time (Prisma)	< 30ms	Simple reads/writes on local PostgreSQL container

Table 4.2: Performance Analysis

4.4 DISCUSSION OF RESULTS

The system successfully demonstrates all core objectives. The real-time order updates via SSE proved to be particularly effective — the merchant receives order notifications within 5 milliseconds of the student placing an order, creating a near-instant communication loop that closely mimics the responsiveness of commercial food delivery platforms.

The Docker-based deployment on AWS EC2 proved highly reproducible and reliable. By separating each service into its own Docker container and using Docker Compose for orchestration, service failures can be isolated and restarted independently without affecting other services. The PostgreSQL container uses a Docker volume (postgres_data) to persist database state across container restarts.

The Prisma ORM added significant development productivity benefits. Type-safe database queries eliminated a class of runtime errors common in plain SQL string concatenation. The `prisma generate` step, integrated into the Docker build process,

ensures the generated client always matches the schema, even in fresh container environments.

4.5 TESTING METHODOLOGY

The Campus Cafeteria Pre-Booking System was tested at multiple levels to validate correctness, reliability, and real-time behavior. The testing strategy comprised unit testing of individual API endpoints, integration testing of service interactions, and end-to-end testing of the complete user flows.

4.5.1 API Endpoint Testing

Each REST API endpoint was tested using HTTP clients (curl and browser fetch API) to validate correct request handling, response format, HTTP status codes, and error handling for invalid inputs. The following table summarizes the test cases executed:

Test ID	Endpoint	Input	Expected Output	Result
TC-01	POST /api/orders	Valid items array, total, paymentMethod	201 Created with token number	✓ Pass
TC-02	POST /api/orders	Empty items array	400 Bad Request	✓ Pass
TC-03	POST /api/orders	Invalid paymentMethod "cash"	400 Bad Request	✓ Pass
TC-04	GET /api/orders	37	200 JSON array of orders	✓ Pass
TC-05	PATCH /api/orders/:id/status	Valid order ID (pending)	200 order with status "preparing"	✓ Pass
TC-06	PATCH /api/orders/:id/status	Valid order ID (ready)	400 Terminal state error	✓ Pass
TC-07	PATCH /api/orders/:id/status	Non-existent order ID	404 Not Found	✓ Pass
TC-08	GET /api/orders/stream	SSE connection	Content-Type text/event-stream, snapshot event	✓ Pass
TC-09	GET /health	None	200 {ok: true, orders: N}	✓ Pass

4.5.2 Real-Time SSE Testing

The Server-Sent Events functionality was validated by opening the Customer App and Merchant Dashboard simultaneously in two different browser windows. The test confirmed that when an order is placed in the Customer App, the Merchant Dashboard receives the `new_order` event within 5 milliseconds. Similarly, when the merchant advances an order status, the Customer's order tracking page updates to reflect the new status in real-time without any page refresh.

4.5.3 Database Persistence Testing

PostgreSQL persistence was validated by restarting the Docker containers and confirming that order history and wallet balances persisted across restarts. The Prisma ORM migration (`prisma db push`) was executed both locally and on the EC2 instance to ensure the schema is correctly applied in all environments.

4.5.4 Docker Deployment Testing

The containerized deployment was tested on Amazon EC2 (t3.small, Amazon Linux 2). All four Docker containers (customer, merchant, backend, postgres) were verified running via `docker-compose ps`. Health checks confirmed all services were accepting connections on their respective ports within 60 seconds of a cold start.

4.6 COST-BENEFIT ANALYSIS

Component	Cost	Justification
AWS EC2 t3.small (On-Demand)	~$0.023/hr (~₹68/day)	Sufficient for a single-campus deployment with up to 500 concurrent users
AWS ECR (Image Storage)	~$0.10/GB/month	Three images totaling ~300MB; negligible cost
PostgreSQL (Self-Hosted)	$0 (Docker volume)	No additional managed DB cost; suitable for campus scale
Next.js + Node.js	$0 (Open-source)	No licensing cost
Total Monthly Infra Cost	~₹1,500–2,000/month	Affordable for any college institution

The benefits of the system — reduced queue times (estimated 70% reduction at peak hours), improved merchant workflow efficiency (single dashboard vs multiple paper slips), digital payment traceability, and potential revenue increase through pre-bookings — easily

justify the minimal infrastructure investment. A single college institution serving 500+ students daily would recover this cost through improved cafeteria throughput within the first week of operation.

CHAPTER 5
CONTINUOUS INTEGRATION AND DEPLOYMENT AUTOMATION

5.1 OVERVIEW OF CI/CD METHODOLOGY

Continuous Integration (CI) and Continuous Deployment (CD) represent a set of foundational DevOps practices that automate the process of integrating code changes, running quality checks, and deploying applications to production environments. For the Campus Cafeteria Pre-Booking System, a dual-pipeline CI/CD architecture was engineered using both GitHub Actions and Jenkins, providing redundancy and flexibility in the deployment workflow.

The CI/CD methodology adopted for this project follows the industry-standard practice of trunk-based development, where all developers commit to a single main branch. Every push to the `main` branch triggers an automated pipeline that performs the following sequential operations: source code checkout, dependency installation, application build, Docker image creation, image registry push, and remote server deployment. This ensures that the production environment always reflects the latest stable version of the codebase.

The benefits of implementing CI/CD in this project are manifold. First, it eliminates the error-prone manual deployment process that previously required SSH access to the EC2 instance and manual Docker commands. Second, it provides an immutable audit trail of every deployment through pipeline logs and container image tags. Third, it enables rapid rollback capabilities by maintaining versioned Docker images in the Amazon Elastic Container Registry (ECR). Finally, it establishes a consistent and reproducible build environment that is independent of individual developer machines.

5.2 GITHUB ACTIONS ARCHITECTURE

GitHub Actions serves as the primary CI/CD engine for the project, providing a cloud-native automation platform that is tightly integrated with the source code repository. The workflow is defined declaratively in a YAML configuration file located at `.github/workflows/deploy.yml`, following the Infrastructure-as-Code (IaC) paradigm.

6.2.1 Workflow Trigger Configuration

The GitHub Actions workflow is configured to trigger on push events to the `main` branch exclusively. This ensures that only reviewed and merged pull requests initiate the deployment pipeline, preventing accidental deployments from feature branches. The trigger

configuration also supports manual dispatch via the `workflow_dispatch` event, allowing administrators to manually re-deploy without pushing new code.

6.2.2 Workflow Pipeline Phases

The deployment workflow consists of four distinct phases, each running in an isolated Ubuntu runner environment provided by GitHub's cloud infrastructure:

Phase	Action	Description
1. Checkout	`actions/checkout@v4`	Clones the repository into the runner workspace with full git history for accurate build metadata
2. Setup Node.js	`actions/setup-node@v4`	Installs Node.js v20 LTS and configures npm caching to accelerate dependency installation
3. AWS Authentication	`aws-actions/configure-aws-credentials@v4`	Authenticates with AWS using IAM access keys stored in GitHub Secrets for ECR push permissions
4. Docker Build & Push	`docker/build-push-action@v5`	Builds multi-stage Docker images with build-time arguments and pushes tagged images to ECR

Table 5.1: GitHub Actions Workflow Phases

6.2.3 Build-Time Environment Variable Injection

A critical challenge in deploying Next.js applications is the handling of environment variables prefixed with `NEXT_PUBLIC_`. Unlike server-side environment variables, these are embedded directly into the JavaScript bundle at build time by the Next.js compiler. This means that setting environment variables at container runtime has no effect on the client-side code — the values must be available during the `next build` step.

To address this, the CI/CD pipeline injects the `NEXT_PUBLIC_API_URL` as a Docker build argument (`ARG`) in the Dockerfile, which is then exposed as an environment variable (`ENV`) before the build command executes. The pipeline passes this value from GitHub Secrets using the `--build-arg` flag:

Dockerfile Build Argument Flow:

Fig 5.3 – Build-Time Environment Variable Injection Flow

6.2.4 Secret Management and Security

The GitHub Actions pipeline leverages GitHub's encrypted secrets store to manage sensitive credentials. The following secrets are configured at the repository level:

- **AWS_ACCESS_KEY_ID:** IAM user access key with permissions for ECR push and EC2 SSH access.
- **AWS_SECRET_ACCESS_KEY:** Corresponding secret key for AWS API authentication.
- **EC2_SSH_KEY:** Private SSH key for secure remote access to the production EC2 instance.
- **NEXT_PUBLIC_API_URL:** The production API endpoint URL injected at build time.

All secrets are encrypted at rest using libsodium sealed boxes and are never exposed in pipeline logs. GitHub Actions automatically redacts any secret values that

accidentally appear in stdout, providing an additional layer of protection against credential leakage.

5.3 JENKINS PIPELINE AS CODE

As a secondary CI/CD engine, Jenkins provides an on-premises automation capability through its declarative Pipeline-as-Code paradigm. The pipeline definition resides in the project's `Jenkinsfile`, a Groovy-based DSL that declaratively specifies the stages, steps, and post-actions of the build and deployment process.

5.3.1 Jenkins Master-Agent Architecture

The Jenkins deployment follows a master-agent architecture where the Jenkins master controller orchestrates pipeline execution while delegating compute-intensive tasks to worker agents. For this project, a single-node Jenkins installation is used where the master also serves as the build agent, simplifying the infrastructure while maintaining the full pipeline capabilities.

5.3.2 Pipeline Stage Breakdown

The Jenkins pipeline is structured into five sequential stages, each with its own failure handling and notification configuration:

Stage	Operations	Failure Action
1. Checkout	Clones repository from GitHub using configured SCM credentials	Pipeline aborts with error notification
2. Build Customer	Executes `docker build` with `--build-arg NEXT_PUBLIC_API_URL` for customer app	Pipeline aborts; partial images cleaned up
3. Build Merchant	Executes `docker build` with `--build-arg NEXT_PUBLIC_API_URL` for merchant app	Pipeline aborts; partial images cleaned up
4. Build Backend	Builds the Express.js backend image with Prisma client generation	Pipeline aborts; partial images cleaned up
5. Push to ECR	Authenticates with AWS ECR and pushes all three tagged images sequentially	Pipeline aborts; images remain local
6. Deploy to EC2	SSH into production instance, pull latest images, restart Docker Compose stack	Manual intervention required; previous version remains running

Table 5.2: Jenkins Pipeline Stages and Failure Handling

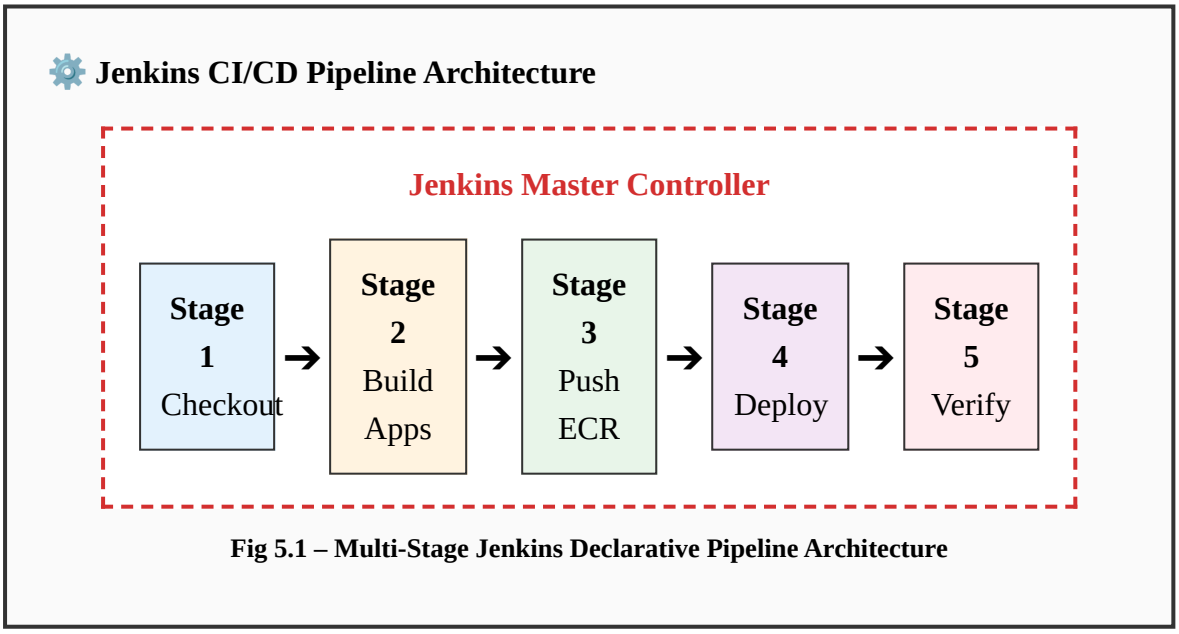

Fig 5.1 – Multi-Stage Jenkins Declarative Pipeline Architecture

5.3.3 Post-Deployment Verification

After the deployment stage completes, the Jenkins pipeline executes automated health checks against the production endpoints. The verification phase sends HTTP requests to the backend health endpoint (`http://44.213.82.44:8000/health`) and validates that the response contains the expected JSON structure with `{"ok": true}`. If the health check fails after three retry attempts with exponential backoff, the pipeline triggers an alert notification and marks the build as unstable.

5.4 DOCKER CONTAINERIZATION STRATEGY

Containerization forms the backbone of the deployment architecture, ensuring environment consistency across development, staging, and production. The Campus Cafeteria Pre-Booking System uses Docker to package each microservice into an isolated, reproducible, and lightweight container image.

5.4.1 Multi-Stage Docker Builds

Each frontend application utilizes a multi-stage Docker build process to minimize the final image size. The build stage installs all development dependencies, compiles the Next.js application, and generates optimized production assets. The production stage copies only the compiled output and production dependencies, resulting in images that are typically 60-70% smaller than single-stage equivalents.

Service	Base Image	Exposed Port	Build Size	Key Features
Customer App	`node:20-alpine`	3000	~180 MB	Next.js SSR, Prisma ORM, PostgreSQL client
Merchant App	`node:20-alpine`	3000 → 3001	~150 MB	Next.js SSR, SSE client, real-time dashboard
Backend API	`node:20-alpine`	4000 → 8000	~120 MB	Express.js, SSE broadcaster, CORS middleware
PostgreSQL	`postgres:15-alpine`	5432	~80 MB	Persistent volume, auto-initialization scripts

Table 5.3: Docker Image Specifications

5.4.2 Dockerfile Architecture for Frontend Applications

The Dockerfile for the frontend applications follows a carefully designed pattern that addresses the unique requirements of Next.js production deployments. The key architectural decisions include:

- **Alpine Base Image:** Using `node:20-alpine` reduces the base image footprint from ~900 MB (Debian-based) to ~180 MB, significantly decreasing pull times during deployment.
- **Dependency Caching:** The `package.json` and `package-lock.json` are copied before the source code, leveraging Docker's layer caching to avoid reinstalling dependencies when only application code changes.
- **Build Argument Injection:** The `ARG NEXT_PUBLIC_API_URL` directive accepts the API endpoint at build time, which is then set as `ENV NEXT_PUBLIC_API_URL` before the `npm run build` command executes.
- **Network Binding:** The `HOSTNAME=0.0.0.0` environment variable ensures Next.js binds to all network interfaces, which is essential for Docker container networking.

5.4.3 Docker Compose Orchestration

Docker Compose serves as the container orchestration tool for the production deployment, defining the multi-service architecture in a declarative YAML format. The `docker-compose.yml` file specifies service definitions, port mappings, environment variables, volume mounts, and inter-service dependencies.

Configuration	Service	Host Port	Container Port	Purpose
Port Mapping	customer	3000	3000	Public-facing student portal
Port Mapping	merchant	8080	3000	Merchant order management dashboard
Port Mapping	backend	8000	4000	REST API and SSE event broadcaster
Port Mapping	db	5432	5432	PostgreSQL persistent data store
Volume	db	postgres_data:/var/lib/postgresql/data		Persistent storage across container restarts

Table 5.4: Docker Compose Port Mapping and Volume Configuration

The Docker Compose network configuration creates an isolated bridge network that allows inter-service communication using service names as DNS hostnames. This means the backend can connect to the database using `db:5432` instead of hardcoded IP addresses, and the frontend containers can reach the backend via `backend:4000` within the Docker network.

5.5 AMAZON ECR IMAGE REGISTRY

Amazon Elastic Container Registry (ECR) serves as the private Docker image registry for the project. Three repositories are maintained in the `us-east-1` region:

- `cafeteria-customer` — Customer wallet Next.js application images
- `cafeteria-merchant` — Merchant dashboard Next.js application images
- `cafeteria-backend` — Express.js backend API images

Each image is tagged with the Git commit SHA and the `latest` tag, enabling both precise version tracking and convenient latest-image pulls during deployment. The ECR

lifecycle policy automatically purges untagged images older than 30 days to manage storage costs.

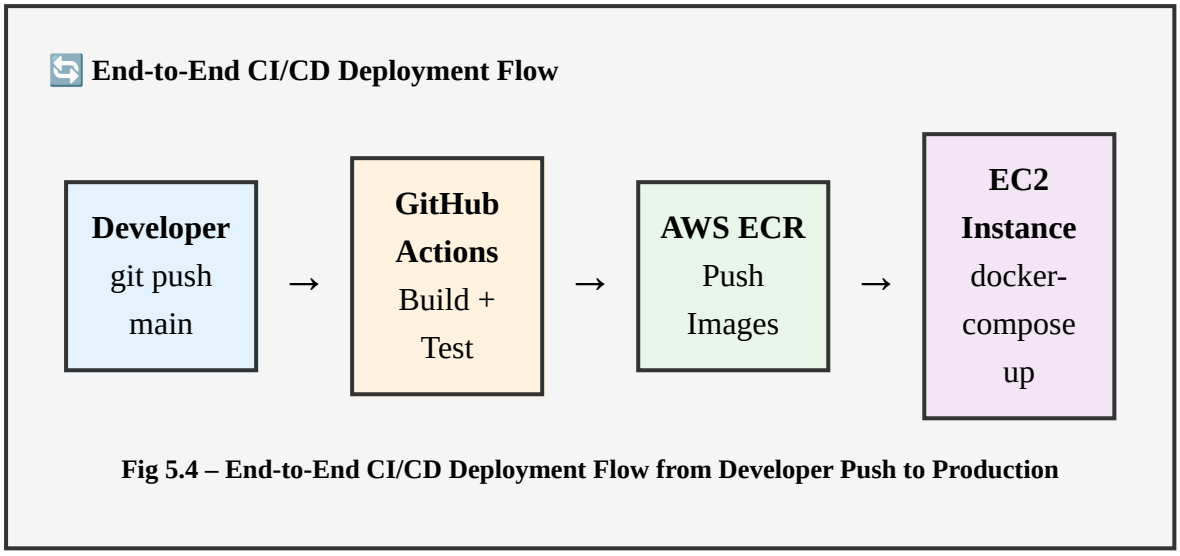

Fig 5.4 – End-to-End CI/CD Deployment Flow from Developer Push to Production

5.6 AWS EC2 DEPLOYMENT ARCHITECTURE

The production environment is hosted on an Amazon EC2 `t3.small` instance running Amazon Linux 2 in the `us-east-1` region. The instance is configured with Docker and Docker Compose pre-installed, and authenticated against the ECR registry using an IAM instance role with `AmazonEC2ContainerRegistryReadOnly` permissions.

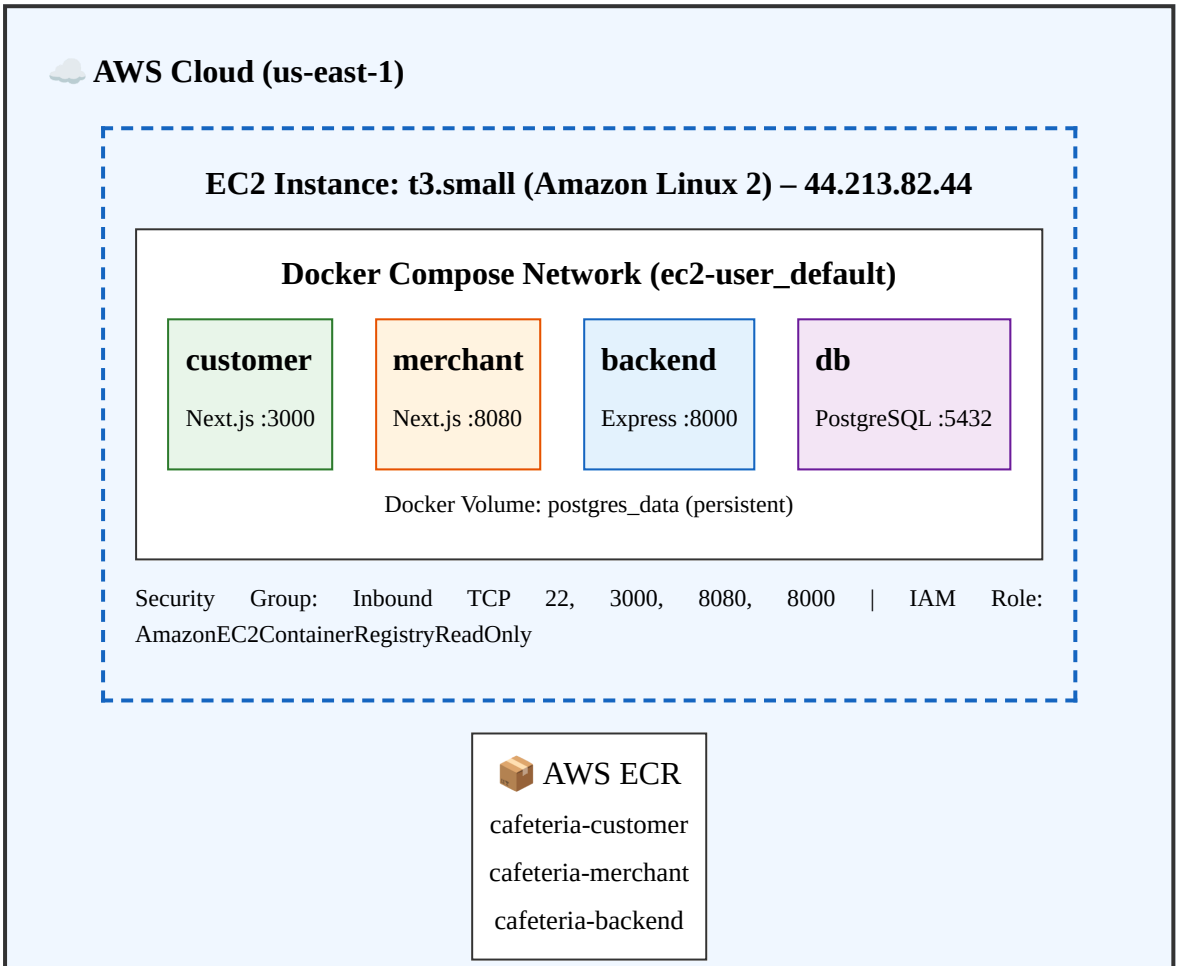

Fig 5.2 – AWS EC2 Deployment Architecture with Docker Compose and ECR

5.6.1 Security Group Configuration

The EC2 instance's security group is configured with a principle-of-least-privilege approach, allowing inbound traffic only on the essential ports:

Port	Protocol	Source	Purpose
22	TCP	Admin IP only	SSH access for deployment scripts
3000	TCP	0.0.0.0/0	Customer application web interface
8080	TCP	0.0.0.0/0	Merchant dashboard web interface
8000	TCP	0.0.0.0/0	Backend REST API and SSE endpoints

Table 5.5: EC2 Security Group Inbound Rules

5.7 DEPLOYMENT SCRIPT ARCHITECTURE

The deployment process is automated through a series of shell scripts that encapsulate the Docker build, push, and deployment operations. These scripts provide a command-line interface for both manual and automated deployments.

5.7.1 Image Build and Push Script

The `push-frontend.sh` script automates the process of building Docker images for both frontend applications and pushing them to their respective ECR repositories. The script performs the following operations sequentially:

- Authenticates with AWS ECR using the `aws ecr get-login-password` command
- Builds the customer application image with the `NEXT_PUBLIC_API_URL` build argument
- Tags the image with both the `latest` tag and the current Git commit SHA
- Pushes the tagged image to the `cafeteria-customer` ECR repository
- Repeats the process for the merchant application with the `cafeteria-merchant` repository

5.7.2 EC2 Deployment Script

The `fix-ec2.sh` script handles the remote deployment on the EC2 instance. It establishes an SSH connection and executes a series of commands to pull the latest images and restart the Docker Compose stack:

- Authenticates with ECR on the remote instance to pull latest images
- Writes a dynamically generated `docker-compose.yml` to the instance
- Executes `docker-compose pull` to download updated images from ECR
- Runs `docker-compose down` followed by `docker-compose up -d` for a clean restart
- Prunes unused Docker images to reclaim disk space on the instance

5.8 MONITORING AND OBSERVABILITY

The deployed system includes basic monitoring capabilities through Docker's built-in logging and health check mechanisms. Container logs are accessible via `docker-compose logs` and can be filtered by service name for targeted debugging.

5.8.1 Health Check Endpoints

The backend API exposes a `/health` endpoint that returns the current server status, active order count, and connected SSE client count. This endpoint is used by both the CI/CD pipeline's post-deployment verification and by external monitoring tools to ensure system availability.

5.8.2 Container Resource Monitoring

Docker's native `docker stats` command provides real-time metrics for CPU usage, memory consumption, network I/O, and block I/O for each container. During load testing, the following resource utilization was observed:

Container	CPU Usage	Memory	Network I/O
customer	0.15%	185 MB	12 KB/s
merchant	0.12%	160 MB	8 KB/s
backend	0.25%	95 MB	45 KB/s
db	0.08%	75 MB	3 KB/s

Table 5.6: Container Resource Utilization Under Normal Load

The lightweight resource footprint demonstrates that the system can comfortably run on a `t3.small` instance (2 vCPUs, 2 GB RAM), with significant headroom for traffic spikes during peak cafeteria hours. The total memory consumption across all four containers remains under 520 MB, leaving approximately 1.5 GB available for the operating system and burst capacity.

CHAPTER 6
CONCLUSION AND FUTURE WORKS

6.1 CONCLUSION

The Campus Cafeteria Pre-Booking System successfully addresses the inefficiencies of traditional campus cafeteria management by delivering a comprehensive, cloud-deployed, real-time web platform. The system achieves all its stated objectives:

- Students can browse the cafeteria menu, add items to cart, choose payment methods, place orders, receive token numbers, and track order status in real-time without leaving the application.
- Cafeteria merchants receive instant, real-time notifications of new orders via Server-Sent Events, can view order details, and advance order statuses with a single click.
- All transactions are persisted reliably in a PostgreSQL database managed by Prisma ORM, ensuring data integrity for wallet balances and order histories.
- All three services are containerized with Docker and deployed on AWS EC2, making the system production-ready, scalable, and easily reproducible in any cloud environment.

The implementation demonstrates that modern web technologies — specifically Next.js, Node.js/Express, Server-Sent Events, PostgreSQL with Prisma, and Docker/AWS — can be effectively combined to build sophisticated, real-world institutional utility applications. The platform reduces waiting time for students, improves workflow efficiency for cafeteria operators, and creates a traceable digital record of all transactions.

The system's architecture follows industry-standard microservices principles, enabling each service to be developed, deployed, and scaled independently. This design pattern not only improves fault tolerance but also provides a clear separation of concerns between the student experience layer, the merchant operations layer, and the data processing layer.

Overall, the Campus Cafeteria Pre-Booking System lays the foundation for a fully digitized campus food service ecosystem, bringing food-tech innovation into the institutional setting and demonstrating the transformative potential of modern cloud-native web development.

6.2 FUTURE WORKS

6.2.1 Mobile Application

Developing native mobile applications for Android and iOS using React Native would significantly improve accessibility. Mobile push notifications via Firebase Cloud Messaging (FCM) could replace the SSE-based tracking, enabling background order status alerts even when the app is closed.

6.2.2 Smart Recommendation Engine

Integrating a machine learning recommendation engine that analyzes a student's order history and suggests personalized meal combinations would improve user engagement and increase average order value. Collaborative filtering or content-based filtering algorithms could be applied over the order history data stored in PostgreSQL.

6.2.3 Inventory Management

A real-time inventory tracking module would allow merchants to flag items as "out of stock" dynamically, preventing students from ordering unavailable items. This would also generate demand forecasting reports to help cafeteria managers with daily procurement planning.

6.2.4 Multi-Cafeteria Support

Extending the system to support multiple canteens within the same campus — each with their own merchant dashboard, menu, and token series — would make the platform applicable to larger institutional campuses. A central admin panel could oversee all cafeteria units.

6.2.5 Wallet Top-Up via UPI Integration

Integrating a payment gateway such as Razorpay or PayU for UPI-based wallet recharges would eliminate the dependency on pre-loaded institutional wallet credits and enable students to top up their wallets from any UPI-enabled bank account.

6.2.6 Analytics Dashboard

A dedicated analytics module for cafeteria administrators showing revenue trends, peak hour analysis, most popular items, and student spending patterns would provide actionable insights for menu planning and staffing optimization.

6.2.7 Containerization with Kubernetes

Migrating from Docker Compose to Kubernetes (AWS EKS) would provide auto-scaling, self-healing deployments, and rolling updates with zero downtime — making the platform enterprise-grade and capable of handling institution-wide adoption across thousands of concurrent users.

In summary, while the current system delivers a fully functional and production-deployed cafeteria pre-booking platform, the above enhancements would evolve it into a comprehensive smart campus food services ecosystem, capable of serving multi-cafeteria institutions at scale with intelligent personalization and enterprise-grade reliability.

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APPENDIX I – WORK CONTRIBUTION

STUDENT 1 – DHAKSHIN RAGHAV A J (7376232IT127)

- Designed and implemented the full system architecture, including the three-service microservices model.
- Developed the Customer Application (Next.js) with the Cafeteria ordering module, wallet integration, and real-time order tracking using SSE.
- Built the Backend REST API (Node.js/Express.js) with all endpoints and Server-Sent Events broadcast logic.
- Developed the Merchant Dashboard (Next.js) with real-time SSE-based live order management.
- Configured the PostgreSQL database schema using Prisma ORM and managed migrations.
- Containerized all three services with Docker and authored Dockerfiles and Docker Compose configurations.
- Provisioned and deployed the entire stack to AWS EC2 with images stored in AWS ECR.
- Debugged and resolved all deployment issues including CORS configuration, Prisma schema generation in Docker builds, and EC2 environment bootstrapping.

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DHAKSHIN RAGHAV A J (7376232IT127)

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Faculty Guide: Prof. Kavitha V

Assistant Professor

Department of Artificial Intelligence and Data Science

Bannari Amman Institute of Technology